

MUHAMMAD KALEEMULLAH KHAN

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EDUCATION

- **PhD in Computer Science**, École de technologie supérieure ÉTS, Montréal, 2022-Continue
Thesis Title: Fair and Secure Federated Adversarial Training.
Supervisor: Prof. Kaiwen Zhang
During my PhD thesis, I proposed a decentralized federated learning framework with an enhanced global model and adaptive aggregator selection for network intrusion detection. I also developed a resource-aware hybrid federated adversarial training model. Furthermore, we introduced an energy-aware adaptive attack model for mobile edge computing. Currently, I am working on designing energy-aware attack strategies for adversarial training.
- **Masters of Science in Computer Science**, COMSATS University Islamabad, 2021
Thesis Title: Efficient Pest Disease identification using Deep Learning
Supervisor: Prof. Dr. Sohail Asghar
During my MS thesis, I have proposed a Light Weight Multi-View CNN for the Plant Disease Identification.
- **Bachelor of Science in Computer Science**, University of Sargodha, 2017
Project Title: Develop an android based application Ease leave

EXPERIENCE

Research Assistant Algoma University, Brampton

August 2025 to December 2025

- Conducted research and development focused on improving energy efficiency in Machine Learning systems.

Lecturer Computer Science

Punjab College, Jhang

June 2021 to November 2021

- Teaching courses like Operating Systems, Theory of Automata, and Introduction to Artificial Intelligence, while mentoring students in refining and developing their Final Year Project ideas.

Machine Learning Research and Development Engineer

SoftBox Technologies, Islamabad

April 2019 to January 2021

- Work on a problem of Image Classification and segmentation, Real-time Object Detection, Face Recognition, Agriculture and Medical Imaging Classification, Facial Emotion Detection

Research Assistant

Department of Computer Science, COMSATS University Islamabad

- Worked on a problem of Breast Cancer Image Classification
- Plant Disease detection and segmentation
- Seizure detection Using EEG signals
- Explainable AI and GAN
- Writing academic papers on achieved findings

RESEARCH INTEREST

Privacy preserving AI, Adversarial Training, Adversarial Attacks, Federated Learning, Distributed ML, Split Learning, Energy Aware ML, Secure and Robust large language models (LLMs), and attack defence mechanisms in distributed ML.

TECHNICAL SKILLS

- **Data Science**: Data Processing, Computer Vision, Machine/Deep Learning, Data Visualization.
- **Programming**: Python(Tensorflow, TensorFlow Federated, OpenCV, Keras, PyTorch, Scikit-learn, Numpy, Pandas, Matplotlib, torchattacks), C and C++.
- **Techniques**: CNN, GAN, LSTM, Autoencoder, Explainable AI, Adversarial Attacks, Transfer Learning, Bayesian Neural Networks, Segmentation, Classification, Clustering, Attention Mechanism, Digital Image Processing and so on.
- **Softwares**: VScode, PyCharm, Sublim Text, Notepad++, Spyder, Latex, Visual Studio, MS Visio, MS Word & Office
- **Others**: Anaconda, Colab, Git, Overleaf, Unity3D, Google ML Kit, Jupyter Notebook.
- **Misc.:** Academic research, teaching, training, consultation, $\LaTeX$ typesetting and publishing.

ACADEMIC PUBLICATIONS

Muhammad KaleemUllah Khan, Kaiwen Zhang, Chamseddine Talhi, and Sayed Muhammad Danish, "A Energy Aware Adaptive Adversarial Attack model for mobile edge computing"[**Status: Under Review**].

Muhammad KaleemUllah Khan, Mohammadreza Sabramooz, Kaiwen Zhang, Chamseddine Talhi, and Sayed Muhammad Danish, "A Resource-based Hybrid Federated Adversarial Training Model with Adaptive Aggregator Selection "[**Status: Under Review**].

Muhammad KaleemUllah Khan "Decentralized Federated Learning with Adaptive Aggregator Selection", in the Proceedings of the 26th International Middleware Conference (MIDDLEWARE-2025).

Muhammad KaleemUllah Khan, Kaiwen Zhang, and Chamseddine Talhi, "FL-EGM: Decentralized Federated Learning using Aggregator Selection with Enhanced Global Model", in the 23rd International Conference of Machine Learning Applications (ICMLA-2024).

Muhammad KaleemUllah Khan "A Light Weight Multi-Features-View Convolution Neural Network For Plant Disease Identification." arXiv preprint arXiv:2605.00903.[Part of Master's Thesis].

Muhammad KaleemUllah Khan, Nadeem Javaid, Shakeeb Murtaza, Maheen Zahid, Wajahat Ali Gilani, and Muhammad Junaid Ali, "Efficient Energy Management Using Fog Computing", in the 21st International Conference on Network-Based Information System (NBIS-2018).

Murtaza, Shakeeb, Nadeem Javaid, **Muhammad KaleemUllah Khan**, Farkhanda Zafar, Rameez Iqbal, and Muhammad Abubaker Hamid. "Efficient Energy Management Assisted by Fog Computing". In Proceedings of the 13th International Conference on Broad-band and Wireless Computing, Communication and Applications (BWCCA-2018).

Gilani, Wajahat Ali, Nadeem Javaid, **Muhammad KaleemUllah Khan**, Hammad Maqbool, Sajid Ali, and Danish Majeed Qureshi. Optimized "Load Balancing Using Cloud Computing" The 21st International Conference on Network-Based Information Systems (NBIS-2018).

Ali, Muhammad Junaid, Nadeem Javaid, Mubariz Rehman, Muhammad Usman Sharif, **Muhammad KaleemUllah Khan**, and Haris Ali Khan. "State Based Load Balancing Algorithm for Smart Grid Energy Management in Fog Computing", The 10th International Conference on Intelligent Networking and Collaborative Systems (INCoS-2018).

Zahid, Maheen, Nadeem Javaid, Kainat Ansar, Kanza Hassan, **Muhammad KaleemUllah Khan**, and Mohammad Waqas. "Hill climbing load balancing algorithm on fog computing". In Proceedings of the 13th International Conference on P2P, Parallel, Grid, Cloud and Internet Computing (3PGCIC-2018).

ACADEMIC PROJECTS

- **Price and Load forecasting:** Semester project of data Mining Course to predict the price and load of electricity using Genetic Algorithms(GA) with Support Vector Machine (SVM).
- **Unity-based game with Turing Machine:**Build a Unity-based Pacman game in the Theory of Computation (ToC) course with different automata machines (PDA, Turing machine) working backend to understand the functionalities of these machines.
- **Name Entity Recognition using Machine learning:**Name Entity (NE) Recognition using different machine learning algorithms as part of the Machine learning course.
- **Enhance the Quality of image:**Semester project for the Neural network Course to enhance the quality of map images using Super Resolution-Generative Adversarial Networks(SR- GAN).
- **Fingerprint Classification:**Semester project for the Computer Vision Course, applying different image processing and deep learning algorithms to classify the fingerprints.
- **Seizure Detection:** Seizure Detection Using Residual Connections CNN models.

OTHER PROJECTS

- **Breast Cancer Classification with Deep Learning**
- **Image Segmentation with Machine Learning**
- **X-rays classification using Deep Learning Models**
- **Cats vs Dogs Classification using CNN Keras**
- **Brain Tumor Detection Using CNN**
- **Heart Disease Prediction Using Machine Learning Techniques**
- **Detecting COVID-19 in X-ray images with Keras, TensorFlow and Deep Learning**

OTHER SKILLS

Software Microsoft Word, Excel, and PowerPoint, Inkscape

Languages English and Urdu

ACADEMIC REFERENCES

- Prof. Kaiwen Zhang
Associate Professor Department of Software Engineering École de technologie supérieure ÉTS, Montréal, Canada
kaiwenzhang@gmail.com
- Prof. Sayed Muhammad Danish
Assistant Professor Department of Computer Science & Technology Algoma University, Brampton, Canada
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